

Awwab Mahdi

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Education

Carleton University **September 2021 - December 2025**
Bachelors of Computer Science Cybersecurity Stream

Experience

Department of National Defence Canada **January 2025 - August 2025**
Data Science Developer 285 Coventry Rd., Ottawa, ON

- Implemented, trained, and tested an optimized 40M parameter document segmentation model(DocStructNet), with 97% accuracy on the DocLayNet and PubLayNet datasets
- Used the AWS fmeval library, along with ECS, EC2, S3 and SageMaker to deploy a government chatbot with an evaluation pipeline
- Reworked a Transport Canada Protected B application to replace AzureBlobService with a local version, and produced an exhaustive report on its functionality and security
- Completed tickets in an Azure DevOps environment, interfacing with Azure GPU VMs, Azure Container Registry, and Azure SQL Database

IRM Consulting & Advisory **December 2023 - January 2024**
AI Developer Remote

- Finetuned an open source LLM from the Hugging Face model library that enabled GPU costs to be reduced by 90%
- Cleaned CIS Controls data from SCF, SOC II, CMMC and other cybersecurity frameworks
- Used LangChain and HuggingFace embeddings to build a file-based chatbot that was able to effectively interact with clients and respond to queries

eWorx Technology **September 2022 - June 2023**
Mobile App Team Colead Remote

- Studied Figma best practices and built several mock-ups to present to the client
- Translated Figma components into functional React Native components
- Used a provided REST API to integrate live data into the app

Projects

Satellite Map 3D Model **January 2025 - March 2025**

- Developed scripts for loading and processing map data from Google Earth Engine
- Created a python Flask server for processing and serving height and texture maps at runtime
- Created a Unity application for displaying a 3D generated world, with multiple perspective controls

Google Earth Engine, Unity, Flask, C#

Profiles

 a--ab.com  [AwwabM](https://github.com/AwwabM)

 [obliviance](https://obliviance.com)

Skills

Programming Languages

Python, Bash, JavaScript, C#, Java, C++, C, SQL

ML Technologies

PyTorch, Keras, TensorFlow, Nvidia Triton Server, LangChain

Web Technologies

HTML/CSS, React.js, React Native, Node.js, MongoDB

Developer Software

Git, Docker, Wireshark, Figma, WordPress, Unity

Other Technologies

Linux, Android, IP/TCP/UDP, Real Time Systems, Agile, Azure DevOps, CI/CD, Cloud Providers(Azure, AWS, GCP)

Publications

DocStructNet: A Foundational Model for Semantic Segmentation of Complex Document Layouts
Preprint (ICPRAM 2026)
September 2025

Languages

English

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French

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Urdu

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